IGCSE 0607 Practice Questions — E3.6 — Combined Questions & Markscheme

Transforming graphs of functions

Total Questions: 18 — **Total Marks:** 110 Marks

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- **E3.6** Transforming graphs of functions
  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(8 Qs)*

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1 A function is defined as $y = f(x)$.

(a) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

..... **[2]**

(b) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Write down the expression for $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[1]**

2 **(a)** The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ to map onto the graph of $y = g(x)$.
Write down $g(x)$ in terms of x.

$g(x) =$ ..... **[1]**

(b) The graph of $y = x^2$ is translated by the vector $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$ to map onto the graph of $y = h(x)$.
Write down $h(x)$ in terms of x.

$h(x) =$ ..... **[1]**

(c) Describe fully the single transformation that maps the graph of $y = x^2$ onto the graph of $y = (x + 4)^2 - 1$.

.....

.....

..... **[2]**

3 A function is defined as $f(x) = 2x - 5$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 7 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in terms of x, giving your answer in its simplest form.

$g(x) =$ ..... **[2]**

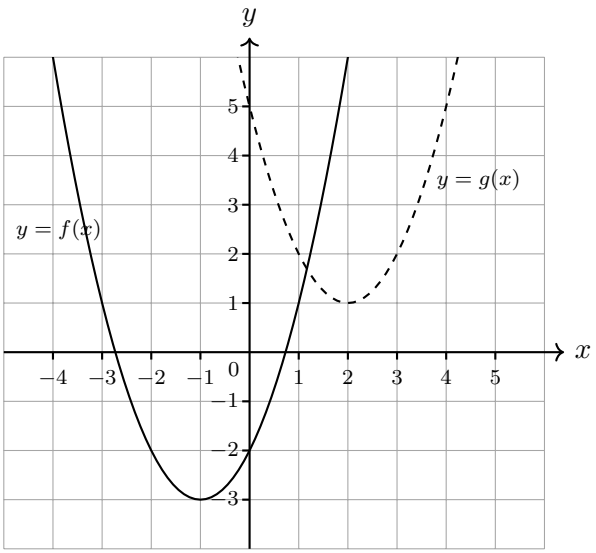
(b) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = f(x - 4)$.

.....

.....

..... **[2]**

4 The diagram shows the graph of $y = f(x)$ and the graph of $y = g(x)$.



(a) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = g(x)$.

.....

.....

..... **[2]**

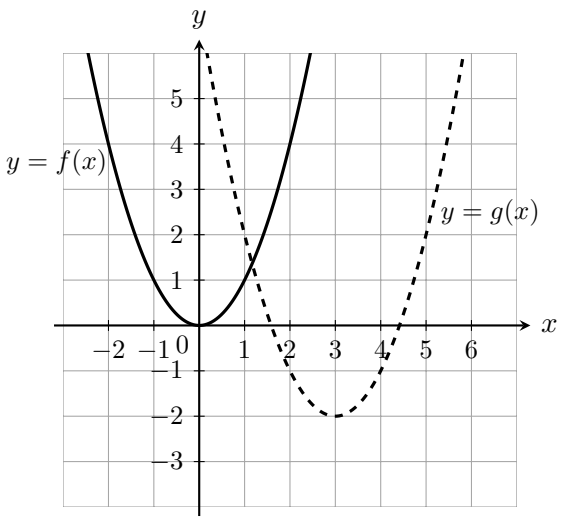
(b) Write $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[2]**

(c) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$.
Write down the coordinates of the vertex of the resulting graph.

(..... , ...) **[1]**

5 The diagram shows the graphs of $y = f(x)$ and $y = g(x)$ on a coordinate grid.



(a) Write down the column vector that describes the translation mapping the graph of $y = f(x)$ onto the graph of $y = g(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Given that $f(x) = x^2$, write down $g(x)$ in the form $(x - a)^2 + b$.

$g(x) =$ ..... **[2]**

(c) Write down the coordinates of the vertex of the graph of $y = f(x + 5) - 1$.

(..... , ...) **[2]**

6 The graph of $y = f(x)$ has a local maximum at the point $P(-2, 5)$ and a local minimum at the point $Q(3, -1)$.

(a) Write down the coordinates of the local maximum of the graph of $y = f(x) - 4$.

(..... , ...) **[1]**

(b) Write down the coordinates of the local minimum of the graph of $y = f(x - 2)$.

(..... , ...) **[1]**

(c) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ such that the image of the local maximum P is at the origin $(0, 0)$.

(i) Find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(ii) Write down the coordinates of the image of the local minimum Q under the same translation.

(..... , ...) **[1]**

7 Consider the function $f(x) = x^2 - 4x + 1$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 2 \\ -5 \end{pmatrix}$ to give the graph of $y = g(x)$.
Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$, where $h(x) = x^2 + 2x - 2$.
Find the translation vector that maps the graph of $y = f(x)$ onto the graph of $y = h(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

8 Let $f(x) = 3x^2$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ to give the graph of $y = g(x)$.

Find $g(x)$ in terms of x.

Give your answer in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

(b) The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 0 \\ -8 \end{pmatrix}$ to give the graph of $y = h(x)$.

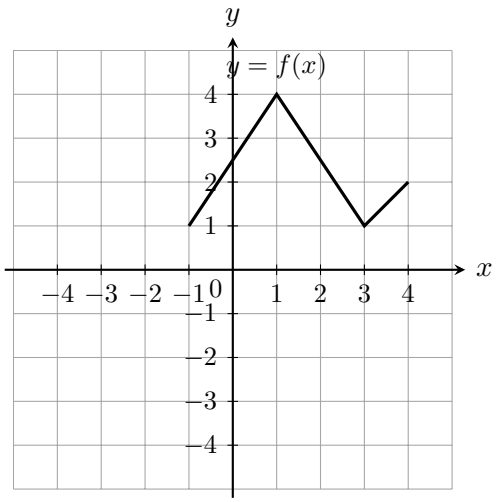
Write down the coordinates of the minimum point of the graph of $y = h(x)$.

(..... , ...) **[2]**

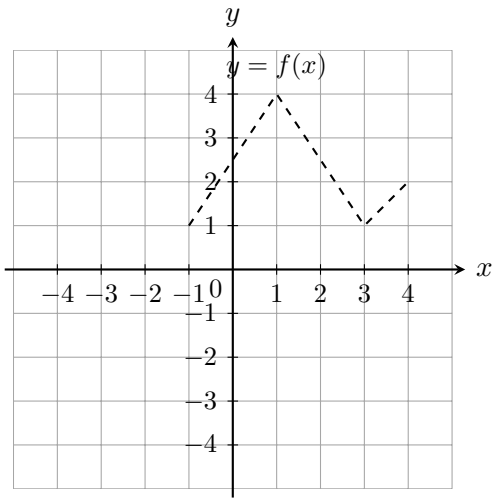
(c) Find the y-intercept of the graph of $y = h(x)$.

..... **[1]**

9 The diagram shows the graph of $y = f(x)$ on a coordinate grid.



(a) On the grid below, draw the graph of $y = f(x + 2) - 3$.



[2]

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a single transformation.
The point $(1, 4)$ on $y = f(x)$ is mapped onto $(1, -1)$ on $y = h(x)$, and the point $(3, 1)$ is mapped onto $(3, -4)$.
Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = h(x)$.

.....

.....

..... **[2]**

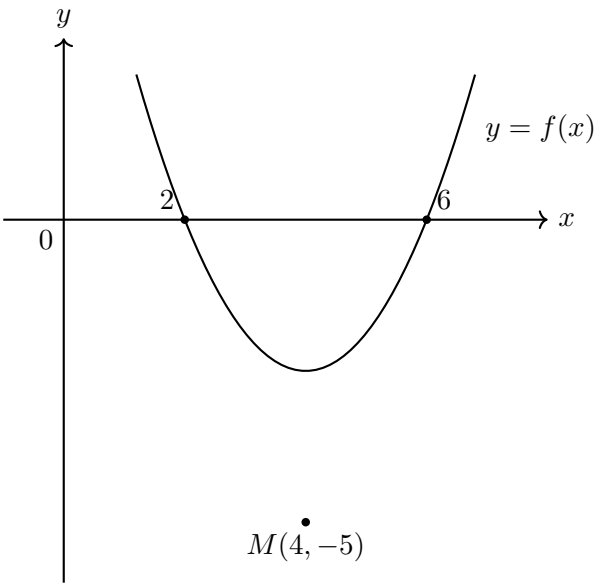
(c) Write down $h(x)$ in terms of $f(x)$.

$h(x) =$ ..... **[1]**

(d) Find the coordinates of the image of the point $(4, 2)$ on $y = f(x)$ when it is mapped onto the graph of $y = f(x - 3) + 4$.

(..... , ...) **[2]**

10 The diagram shows a sketch of the graph of $y = f(x)$.
The curve intersects the x-axis at $(2, 0)$ and $(6, 0)$ and has a minimum point at $M(4, -5)$.



(a) A transformation maps the graph of $y = f(x)$ onto the graph of $y = f(x) + 4$.
Write down the coordinates of the image of point M.

((..... , ...)) **[1]**

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$.
Write down the expression for $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[2]**

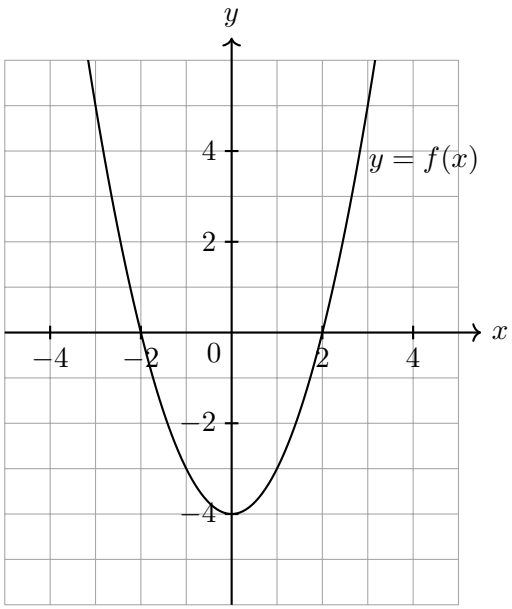
(c) The graph of $y = f(x - k)$ passes through the origin $(0, 0)$.
Find the two possible values of the constant k.

$k =$ ..... , $k =$ ..... **[2]**

(d) The function $h(x)$ is such that the graph of $y = h(x)$ has a minimum point at $(7, -5)$.
Find $h(x)$ in terms of $f(x)$.

$h(x) =$ ..... **[1]**

11 The diagram shows the graph of $y = f(x)$, where $f(x) = x^2 - 4$.



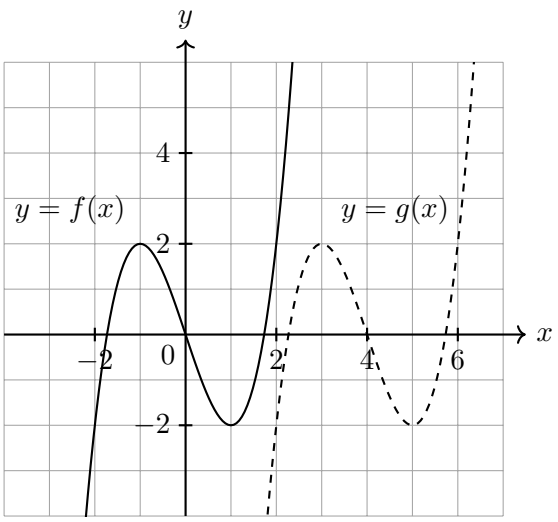
(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ to give the graph of $y = g(x)$.
Write down the expression for $g(x)$ in terms of x, simplifying your answer.

$g(x) =$ ..... **[2]**

(b) Write down the coordinates of the turning point of the graph of $y = f(x + 3)$.

(..... , ...) **[2]**

12 The diagram shows the graph of $y = f(x)$ and its image $y = g(x)$ after a single transformation.



(a) Describe fully the single transformation that maps the graph of $y = f(x)$ onto the graph of $y = g(x)$.

.....

.....

..... **[2]**

(b) Given that $f(x) = x^3 - 3x$, find $g(x)$ in terms of x, giving your answer in expanded and simplified form.

$g(x) =$ ..... **[4]**

13 A function is given as $f(x) = x^2 - 2x - 3$.

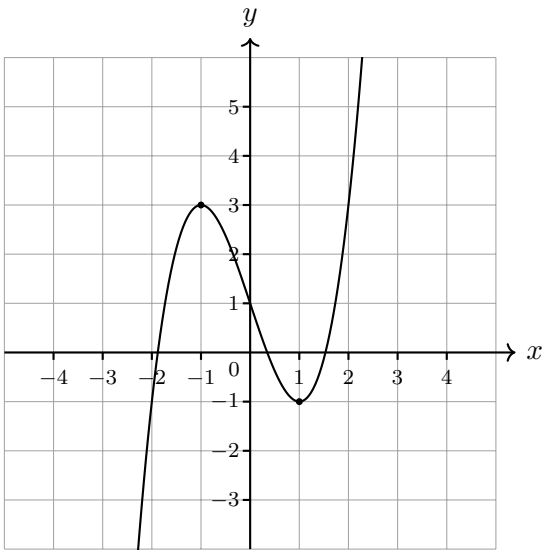
(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$ to form the graph of $y = g(x)$.
Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[4]**

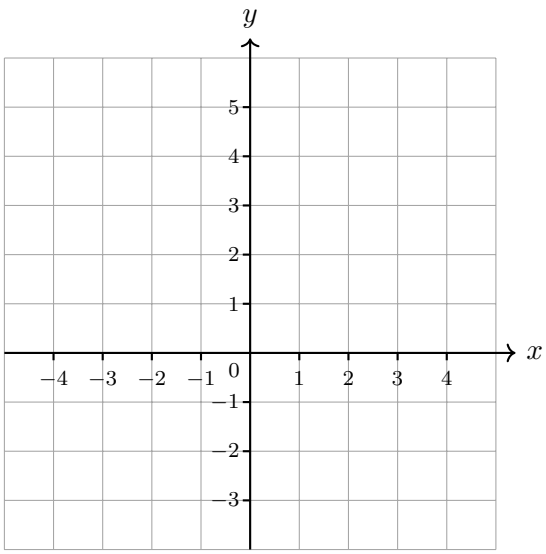
(b) The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$.
Solve the equation $h(x) = 0$.

$x =$ ..... , $x =$ ..... **[3]**

14 The diagram shows the graph of $y = f(x)$ for $-4 \leq x \leq 4$.
The curve has a local maximum at $(-1, 3)$, a local minimum at $(1, -1)$, and passes through the origin $(0, 1)$.



(a) On the grid, sketch the graph of $y = f(x - 2) + 1$.



[3]

(b) Given that $f(x) = x^3 - 3x + 1$, find the coordinates of the local minimum point on the graph of $y = f(x + k) - 2$, in terms of k.

(..... , ...) **[2]**

(c) The graph of $y = f(x) + p$ has exactly two x-intercepts.
Write down the two possible values of p.

$p =$ ..... , $p =$ ..... **[2]**

15 The function f is defined by $f(x) = 2x^2 - 8x + 11$.

(a) Write $f(x)$ in the form $a(x + h)^2 + k$.

$f(x) =$ ..... **[3]**

(b) The graph of $y = f(x)$ is translated to give the graph of $y = g(x)$, where $g(x) = 2x^2 + 4x - 3$.
Find the translation vector that maps the graph of $y = f(x)$ onto the graph of $y = g(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[4]**

16 The functions $f(x)$ and $g(x)$ are defined as $f(x) = x^2 - 6x + 14$ and $g(x) = \dfrac{3}{x+2} - 5$.

(a) By writing $f(x)$ in the form $(x-a)^2 + b$, find the column vector of the translation that maps the graph of $y = x^2$ onto the graph of $y = f(x)$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

(b) The graph of $y = g(x)$ is translated by the vector $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ to map onto the graph of $y = h(x)$.

Find $h(x)$ in its simplest form.

$h(x) =$ ..... **[3]**

(c) Write down the equations of the vertical and horizontal asymptotes of the graph of $y = h(x)$.

$x =$ ..... , $y =$ ..... **[2]**

17 A function is defined as $f(x) = \dfrac{20}{x}$.

(a) The graph of $y = f(x)$ is mapped onto the graph of $y = g(x)$ by a translation with vector $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$.

(i) Write down an expression for $g(x)$.

$g(x) =$ ..... **[1]**

(ii) Solve the equation $g(x) = 0$.

$x =$ ..... **[2]**

(b) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ to give the graph of $y = h(x)$.
The graph of $y = h(x)$ passes through the point $(8, 5)$.
Work out the value of k.

$k =$ ..... **[3]**

(c) Let $p(x) = f(x - 3) + 7$.
Write down the equations of the vertical and horizontal asymptotes of the graph of $y = p(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

18 The function f is defined by $f(x) = 2x^2 - 5x + 1$.

(a) The graph of $y = f(x)$ is translated by the vector $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$ to give the graph of $y = g(x)$.

(i) Write down $g(x)$ in terms of $f(x)$.

$g(x) =$ ..... **[1]**

(ii) Find $g(x)$ in the form $ax^2 + bx + c$.

$g(x) =$ ..... **[3]**

(b) The graph of $y = f(x)$ is mapped onto the graph of $y = h(x)$ by a single translation, where $h(x) = 2x^2 - 13x + 18$.
Find the column vector of this translation.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[4]**

(c) Find the coordinates of the point of intersection of the graphs $y = f(x)$ and $y = g(x)$.

(..... , ...) **[3]**

1 **(a)** Translation **[B1]**

by vector $\begin{pmatrix}4\\0\end{pmatrix}$ **[B1]**

(b) $f(x)+7$ **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** x^2-5 **[B1]**

(b) $(x-3)^2$ **[B1]**

(c) Translation **[B1]**

by vector $\begin{pmatrix}-4\\-1\end{pmatrix}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** $g(x)=f(x)+7$ **[M1]**

$g(x)=2x+2$ **[A1]**

(b) Translation **[B1]**

by vector $\begin{pmatrix}4\\0\end{pmatrix}$ **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Translation **[B1]**

By vector $\begin{pmatrix}3\\4\end{pmatrix}$ **[B1]**

(b) $f(x-3)+4$ **[B2]**

(Award [B1] for $f(x-3)$ or $+4$ seen)

(c) $(-3,-3)$ **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** Horizontal shift 3 seen **[B1]**

$\begin{pmatrix}3\\-2\end{pmatrix}$ **[B1]**

(b) $(x-3)^2$ seen **[B1]**

$(x-3)^2-2$ **[B1]**

(c) $x=-5$ **[B1]**

$(-5,-1)$ **[B1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** $(-2, 1)$ **[B1]**

(b) $(5, -1)$ **[B1]**

(c) **(i)** $a = 2$ **[B1]**

$b = -5$ **[B1]**

(ii) $(5, -6)$ **[B1]**

(Strict follow through from their vector in (c)(i) applied to $(3, -1)$)

[Back to Question] *[Total Marks: 5]*

7 **(a)** Writes $g(x) = f(x-2) - 5$ or $(x-2)^2 - 4(x-2) + 1 - 5$ **[M1]**

Expands correctly: $(x^2 - 4x + 4) - 4x + 8 - 4$ **[M1]**

$x^2 - 8x + 8$ **[A1]**

(b) Completes the square: $f(x) = (x-2)^2 - 3$ and $h(x) = (x+1)^2 - 3$ **[M1]**

Identifies horizontal shift of -3 units **[A1]**

$\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** $g(x) = f(x+2) = 3(x+2)^2$ **[M1]**

$3(x^2 + 4x + 4)$ **[M1]**

$3x^2 + 12x + 12$ **[A1]**

(b) $(-2, -8)$ **[B2]**

(Allow B1 for each correct coordinate)

(c) Substitutes $x = 0$ into $h(x) = 3(x+2)^2 - 8$ (or similar valid method) **[M1]**

4 **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** Translation by 2 units left or 3 units down **[M1]**

Correct line segments joining $(-3, -2)$, $(-1, 1)$, $(1, -2)$, and $(2, -1)$ **[A1]**

(b) Translation **[B1]**

Vector $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$ **[B1]**

(c) $f(x) - 5$ **[B1]**

(d) $x = 4 + 3 = 7$ or $y = 2 + 4 = 6$ **[M1]**

$(7, 6)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** $(4, -1)$ **[B1]**

(b) $f(x+3)+2$ **[B2]**

(B1 for $f(x+3)$ and B1 for $+2$)

(c) $f(-k) = 0$, so $-k = 2$ or $-k = 6$ **[M1]**

$k = -2$ and $k = -6$ **[A1]**

(d) $f(x-3)$ **[B1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** Correct vertical shift applied: $(x^2-4)+5$ **[M1]**

x^2+1 **[A1]**

(b) $(-3, -4)$ **[B2]**

If 0 scored, award 1 mark for x-coordinate $= -3$ or y-coordinate $= -4$ **[B1]**

[Back to Question] *[Total Marks: 4]*

12 **(a)** Translation **[B1]**

by vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ **[B1]**

(b) Recognising that $g(x) = f(x-4)$ and substituting: $(x-4)^3 - 3(x-4)$ **[M1]**

Expanding $(x-4)^3 = x^3 - 12x^2 + 48x - 64$ **[B2]**

If 0 scored for cubic expansion, award 1 mark for $(x-4)(x^2-8x+16)$ **[M1]**

$x^3 - 12x^2 + 45x - 52$ **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** $g(x) = f(x+4)$ **[M1]**

$(x+4)^2 - 2(x+4) - 3$ **[M1]**

$x^2 + 8x + 16 - 2x - 8 - 3$ **[M1]**

$x^2 + 6x + 5$ **[A1]**

(b) $h(x) = x^2 - 2x - 3 - 5 = x^2 - 2x - 8$ **[M1]**

$(x-4)(x+2) = 0$ **[M1]**

$x = 4$ and $x = -2$ **[A1]**

[Back to Question] *[Total Marks: 7]*

14 **(a)** Local maximum plotted at $(1, 4)$ **[B1]**

Local minimum plotted at $(3, 0)$ and y-intercept or key point $(2, 2)$ correct **[B1]**

Smooth cubic curve drawn through translated points with correct shape **[B1]**

(b) x-coordinate $= 1 - k$ **[B1]**

y-coordinate $= -1 - 2 = -3$, giving $(1 - k, -3)$ **[B1]**

(c) $p = -3$ **[B1]**

$p = 1$ **[B1]**

[Back to Question] *[Total Marks: 7]*

15 (a) $2(x^2 - 4x) + 11$ or $a = 2$ **[B1]**

$2[(x - 2)^2 - 4] + 11$ or $h = -2$ **[B1]**

$2(x - 2)^2 + 3$ **[A1]**

(b) Completing the square for $g(x)$: $2(x + 1)^2 - 5$ OR equating $f(x - p) + q = 2x^2 + 4x - 3$ **[M1]**

Identifying the minimum point of $f(x)$ as $(2, 3)$ and the minimum point of $g(x)$ as $(-1, -5)$ **[M1]**

Horizontal shift $= -1 - 2 = -3$ **[A1]**

Vertical shift $= -5 - 3 = -8$, giving vector $\begin{pmatrix} -3 \\ -8 \end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 (a) Finds $(x - 3)^2$ as part of completing the square: $f(x) = (x - 3)^2 - 9 + 14$ **[M1]**

$f(x) = (x - 3)^2 + 5$ **[M1]**

Translation vector $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ **[A1]**

(b) Substitutes $x - 4$ into $g(x)$: $g(x - 4) = \dfrac{3}{(x - 4) + 2} - 5$ **[M1]**

Adds 2 to the function: $h(x) = \dfrac{3}{x - 2} - 5 + 2$ **[M1]**

$h(x) = \dfrac{3}{x - 2} - 3$ or equivalent single fraction $\dfrac{9 - 3x}{x - 2}$ **[A1]**

(c) $x = 2$ **[B1]**

$y = -3$ **[B1]**

(Strict follow through from their $h(x)$ in part (b))

[Back to Question] *[Total Marks: 8]*

17 (a) (i) $\dfrac{20}{x} - 4$ **[B1]**

(ii) $\dfrac{20}{x} - 4 = 0$ **[M1]**

$x = 5$ **[A1]**

(b) $h(x) = f(x - k) = \dfrac{20}{x - k}$ **[M1]**

$\frac{20}{8-k} = 5$ **[M1]**

$k = 4$ **[A1]**

(c) Vertical asymptote: $x = 3$ **[B1]**

Horizontal asymptote: $y = 7$ **[B1]**

[Back to Question] *[Total Marks: 8]*

18 (a) (i) $f(x+2) - 6$ **[B1]**

(ii) $2(x+2)^2 - 5(x+2) + 1 - 6$ **[M1]**

$2(x^2 + 4x + 4) - 5x - 15$ **[M1]**

$2x^2 + 3x - 7$ **[A1]**

(b) $2(x-p)^2 - 5(x-p) + 1 + q = 2x^2 - 13x + 18$ **[M1]**

$-4p - 5 = -13$ giving $p = 2$ **[A1]**

$2(2)^2 + 5(2) + 1 + q = 18$ giving $q = -1$ **[M1]**

$\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ **[A1]**

(c) $2x^2 - 5x + 1 = 2x^2 + 3x - 7$ **[M1]**

$8x = 8$ giving $x = 1$ **[A1]**

$(1, -2)$ **[A1]**

[Back to Question] *[Total Marks: 11]*